

Lesson 5 · Timed practice

NCEA Level 2 · AS 91262 · Mixed questions

Time: 30 minutes

Answer version · tutor's working, not an official mark scheme

Exam conditions: no notes; calculator allowed. Show all working and write the derivative first. [A] Achieved, [M] Merit, [E] Excellence.

1. [A]

- (a) Find the gradient of $f(x) = x^4 - 3x^2 + 2x$ at the point where $x = -1$.

$$f'(x) = 4x^3 - 6x + 2; f'(-1) = -4 + 6 + 2 = 4$$

- (b) Find the equation of the tangent to $y = 2x^2 - 5x + 1$ at $x = 3$.

$$\text{Point: } y = 18 - 15 + 1 = 4; \frac{dy}{dx} = 4x - 5, \text{ so } m = 7$$

$$y - 4 = 7(x - 3), \text{ so } y = 7x - 17$$

- (c) $f'(x) = 3x^2 - 8x + 2$ and $f(2) = 1$. Find $f(x)$.

$$f'(x) = x^3 - 4x^2 + 2x + c; 8 - 16 + 4 + c = 1, \text{ so } c = 5$$

$$f(x) = x^3 - 4x^2 + 2x + 5$$

- (d) For which values of x is $f(x) = x^3 - 6x^2 + 5$ decreasing?

$$f'(x) = 3x^2 - 12x = 3x(x - 4) < 0$$

$$\text{Negative between the roots: } 0 < x < 4$$

2. [M]

- (a) A particle moves with $s = 4t^3 - 15t^2 + 12t$ metres after t seconds. Find the times when it is at rest.

$$v = 12t^2 - 30t + 12 = 6(2t^2 - 5t + 2) = 6(2t - 1)(t - 2)$$

$$\text{At rest at } t = 0.5 \text{ s and } t = 2 \text{ s}$$

- (b) A drone takes off from the ground with vertical velocity $v = 6t - t^2 \text{ m s}^{-1}$ for $0 \leq t \leq 6$. Find its maximum height, and justify that it is the maximum.

$$s = 3t^2 - \frac{1}{3}t^3 \text{ (from the ground, } c = 0)$$

$$v = t(6 - t) \geq 0 \text{ for } 0 \leq t \leq 6, \text{ so the drone rises the whole time;}$$

$$v = 0 \text{ at } t = 6: \text{ maximum height } s(6) = 108 - 72 = 36 \text{ m}$$

Question 3 [E]

- (iii) At another match, the number of spectators is modelled by the equation $P = kt^2 - 2t^4 + 750$, where k is a constant.

The number of spectators is growing fastest when $t = 4$.

Find the value of k .

Explain your reasoning clearly, using correct mathematical statements.

Rate of change of spectators: $P' = 2kt - 8t^3$

Growing fastest when P' is a maximum, so $P'' = 2k - 24t^2 = 0$

At $t = 4$: $2k - 24(16) = 0$, so $k = 192$

Check it is a maximum of P' : $P''' = -48t = -192 < 0$

and $P'(4) = 2(192)(4) - 8(64) = 1536 - 512 = 1024 > 0$, so the crowd is growing

$k = 192$